COMP1870 Theoretical Foundations of Computer Science I

School of Computer Science, University of Leeds

Worksheet - week 7

This worksheet contains a combination of *formative activities* (which contribute towards your learning) and *summative activities* (which you will complete and submit to be assessed as part of your portfolio).

- Exercises marked with red borders are summative and must be submitted in Gradescope.
- Activities marked by (*) are advanced, and may take some time to complete.

Expectations:

1. **Timeliness** You should complete all of the activities in the order provided and submit your portfolio evidence on PebblePad before the deadline.
2. **Presentation:** You should present all of your work clearly and concisely following any additional guidance provided by the module staff.
3. **Integrity:** You are responsible for ensuring that the evidence you submit as part of your portfolio is **entirely your own work**. You can find out more about Academic integrity on the Skill@library website. All work you submit for assessment is subject to the academic integrity policy.

Feedback: Feedback on formative activities will be provided via Lab classes and tutorials. Feedback on evidence submitted as part of the portfolio will be available on Gradescope.

Support opportunities: Support with the activity sheet is available in tutorials and the Math Lunch Hour. Individual support is available via office hours.

Expected completion date: Friday 20 November 2026

All portfolio tasks are rated AMBER by the university: AI tools can be used in an assistive role. You are permitted to use AI tools for specific defined processes within the assessment. Within this assessment you may use Generative AI to:

- help you start to understand more complex ideas by providing accessible summaries.
- test your knowledge against a piece of content.
- help identify and correct spelling mistakes and grammatical errors in your work.
- provide feedback and advice on your overall coding style.

You must not use Gen AI to:

- produce written content (code, comments or any other content) which you then submit as your work, regardless of whether you make changes to it.
- rewrite or make changes to any of your work.

Learning outcomes

On completion of this activity sheet you will have:

1. demonstrated an understanding of basic operations on matrices and their properties.
2. ability to write a system of linear equations in matrix format.

Questions

1. A matrix is a **scalar matrix** if it is a diagonal matrix and all the diagonal entries are equal.
   Let A be an $n \times n$ matrix and B be a $n \times n$ scalar matrix. Then, prove that $AB = BA$.

2. Write the following system of linear equation in matrix format.

$$\begin{aligned} x_1 + 3x_3 &= 7 \\ x_2 + 5x_4 &= 8 \\ x_1 + x_3 &= x_2 \\ x_1 + 7x_2 + 2 &= 0 \\ 6x_3 + 9x_4 &= 5 \end{aligned}$$

3. Let A and B be two $m \times n$ matrices and let C be an $n \times p$ matrix. Prove that $(A + B)C = BC + AC$.

4. Let A be an unknown 4×4 matrix. Let D_1 and D_2 be two matrices defined below.

$$D_1 = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad D_2 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

   We also know that

$$D_1A = \begin{bmatrix} 3 & 6 & 9 & 12 \\ 12 & 15 & 18 & 21 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \qquad D_2A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 16 & 18 & 20 & 22 \\ 22 & 24 & 26 & 28 \end{bmatrix}$$

   Compute $6A$.

5. Let $A = [1, 2, 3]$ be $B = [4, 3, 2, 1]$ be two matrices.

   (a) Is $A^T + B$ defined (i,e., A^T and B are conformable for addition)? If so, then write $A^T + B$. Otherwise explain why it is not defined.

   (b) Is A^TB defined? If so, then write A^TB. Otherwise explain why it is not defined.

   (c) Is AB^T defined? If so, then write AB^T. Otherwise explain why it is not defined.